

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

|              |                                  |
|--------------|----------------------------------|
| Product Name | <u>Sulfuric acid, 20% fuming</u> |
| CAS No       | 8014-95-7                        |
| Synonyms     | Oleum                            |

|                         |                                                                                                      |
|-------------------------|------------------------------------------------------------------------------------------------------|
| Product Code            | <b>33274</b>                                                                                         |
| Address                 | ThermoFisher Scientific Australia Pty Ltd<br>5 Caribbean Drive, Scoresby<br>VICTORIA 3179, Australia |
| Emergency Tel.          | <b>CHEMTREC®</b><br><b>03 9757 4559 or +613 9757 4559</b>                                            |
| Telephone / Fax Numbers | Tel: 1300 735 292<br>Fax: 1800 067 639                                                               |
| E-mail address          | ANZinfo@thermofisher.com                                                                             |

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

**Physical hazards**  
No hazards identified

**Health hazards**

Acute Inhalation Toxicity - Vapors  
Skin Corrosion/Irritation  
Serious Eye Damage/Eye Irritation  
Specific target organ toxicity - (single exposure)

Category 2  
Category 1 A  
Category 1  
Category 3

**Environmental hazards**  
No hazards identified

### Label Elements



Skull and Crossbones



Corrosion

**Signal Word****Danger****Hazard Statements**

H314 - Causes severe skin burns and eye damage

H330 - Fatal if inhaled

H335 - May cause respiratory irritation

AUH014 - Reacts violently with water

**Precautionary Statements**

P260 - Do not breathe dust/fume/gas/mist/vapors/spray

P201 - Obtain special instructions before use

P202 - Do not handle until all safety precautions have been read and understood

P264 - Wash face, hands and any exposed skin thoroughly after handling

P271 - Use only outdoors or in a well-ventilated area

P284 - Wear respiratory protection

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P363 - Wash contaminated clothing before reuse

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P310 - Immediately call a POISON CENTER or doctor

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

P405 - Store locked up

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information****Section 3 - Composition and Information on Ingredients**

| Component             | CAS No    | Weight % |
|-----------------------|-----------|----------|
| Sulfuric acid, fuming | 8014-95-7 | 80       |
| Sulfur trioxide       | 7446-11-9 | 20       |

**Section 4 - First Aid Measures****Inhalation**

Remove to fresh air. If breathing is difficult, give oxygen. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

|                                            |                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                         | Immediate medical attention is required. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.                                                                                                                                                                 |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                  |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                              |
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes. . Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                                                                                            |

## Section 5 - Fire Fighting Measures

**Suitable Extinguishing Media**  
Carbon dioxide (CO<sub>2</sub>). Dry chemical.

**Extinguishing media which must not be used for safety reasons**  
Water.

**Hazardous Decomposition Products**  
Sulfur oxides.

**Specific Hazards Arising from the Chemical**  
Contact with water liberates toxic gas. Water reactive. Produce flammable gases on contact with water.

**Special protective equipment and precautions for fire fighters**  
As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

**Emergency procedures**  
Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Use personal protective equipment as required. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Do not use metal tools or equipment. DO NOT GET WATER on spilled substance or inside containers

**Environmental Precautions**  
Should not be released into the environment. Prevent product from entering drains. Keep out of waterways. See Section 12 for additional Ecological Information.

### Methods for Containment and Clean Up

**Clean-up methods - small spillage**  
Use personal protective equipment as required. Provide adequate ventilation. Neutralize with lime milk or soda and flush with plenty of water. Soak up with inert absorbent material. Sweep up and shovel into suitable containers for disposal. Keep from any possible contact with water, because of violent reaction and possible flash fire. Do not flush into surface water or sanitary sewer system.

**Clean-up methods - large spillage**  
Typically only supplied in small quantities as packaged goods.  
If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Do not allow contact with water because of violent reaction. Handle under inert gas, protect from moisture.

**Conditions for Safe Storage, Including any Incompatibilities**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Protect from moisture. Corrosives area.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

**Exposure limits**

The product does not contain any hazardous materials with occupational exposure limits established.

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Exposure Controls****Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Butyl rubber   | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection**

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:**

Particulates filter conforming to EN 143 Acid gases filter Type E Yellow conforming to EN14387 (or AUS/NZ equivalent)

**Recommended half mask:-**

Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted

|                                 |                                                                        |
|---------------------------------|------------------------------------------------------------------------|
| Hygiene Measures                | Handle in accordance with good industrial hygiene and safety practice. |
| Environmental exposure controls | No information available.                                              |

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                         |                          |                                          |
|-----------------------------------------|--------------------------|------------------------------------------|
| Appearance                              | Light brown              |                                          |
| Physical State                          | Liquid                   |                                          |
| Odor                                    | pungent                  |                                          |
| Odor Threshold                          | No data available        |                                          |
| pH                                      | No information available |                                          |
| Melting Point/Range                     | 2 °C / 35.6 °F           |                                          |
| Softening Point                         | No data available        |                                          |
| Boiling Point/Range                     | 138 °C / 280.4 °F        | @ 760 mmHg                               |
| Flash Point                             | No information available | <b>Method -</b> No information available |
| Evaporation Rate                        | No data available        |                                          |
| Flammability (solid,gas)                | Not applicable           | Liquid                                   |
| Explosion Limits                        | No data available        |                                          |
| Vapor Pressure                          | 1 mmHg @ 146 °C          |                                          |
| Vapor Density                           | 3 (Air = 1.0)            | (Air = 1.0)                              |
| Specific Gravity / Density              | 1.920                    |                                          |
| Bulk Density                            | Not applicable           | Liquid                                   |
| Water Solubility                        | Miscible                 |                                          |
| Solubility in other solvents            | No information available |                                          |
| Partition Coefficient (n-octanol/water) |                          |                                          |
| Autoignition Temperature                | No data available        |                                          |
| Decomposition Temperature               | No data available        |                                          |
| Viscosity                               | No data available        |                                          |
| Explosive Properties                    | No information available |                                          |
| Oxidizing Properties                    | No information available |                                          |
| <b>Other information</b>                |                          |                                          |
| Molecular Formula                       | H2 O4 S . S O3           |                                          |
| Molecular Weight                        | 178.14                   |                                          |

## Section 10 - Stability and Reactivity

|                                  |                                                                                        |
|----------------------------------|----------------------------------------------------------------------------------------|
| Reactivity                       | Yes                                                                                    |
| Stability                        | Hygroscopic. Reacts violently with water, liberating extremely flammable gases.        |
| Conditions to Avoid              | Incompatible products, Exposure to moist air or water.                                 |
| Incompatible Materials           | Bases, Strong oxidizing agents, Ammonia, Combustible material, Metals, Reducing Agent. |
| Hazardous Decomposition Products | Sulfur oxides.                                                                         |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                               |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

**Product Information****(a) acute toxicity;****Oral**

Based on available data, the classification criteria are not met

**Dermal**

No data available

**Inhalation**

Category 2

**Toxicology data for the components**

| Component             | LD50 Oral                 | LD50 Dermal | LC50 Inhalation                                                                        |
|-----------------------|---------------------------|-------------|----------------------------------------------------------------------------------------|
| Sulfuric acid, fuming | LD50 = 2140 mg/kg ( Rat ) |             | LC50 = 347 ppm ( Rat ) 1 h                                                             |
| Sulfur trioxide       |                           |             | LC50 = 1136 mg/m <sup>3</sup> ( Rat ) 1 h<br>LC50 = 1375 mg/m <sup>3</sup> ( Rat ) 1 h |

**(b) skin corrosion/irritation;** Category 1 A**(c) serious eye damage/irritation;** Category 1**(d) respiratory or skin sensitization;****Respiratory**

No data available

**Skin**

No data available

**(e) germ cell mutagenicity;** No data available**(f) carcinogenicity;** No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen.  
Exposure to strong inorganic mists containing sulfuric acid may cause cancer by inhalation.

| Component             | Australia | New Zealand | New South Wales | Western Australia | IARC    | EU | UK | Germany |
|-----------------------|-----------|-------------|-----------------|-------------------|---------|----|----|---------|
| Sulfuric acid, fuming |           |             |                 |                   | Group 1 |    |    |         |
| Sulfur trioxide       |           |             |                 |                   | Group 1 |    |    |         |

**(g) reproductive toxicity;** No data available**(h) STOT-single exposure;** Category 3**Results / Target organs**

Respiratory system

**(i) STOT-repeated exposure;** No data available**Target Organs**

No information available.

**(j) aspiration hazard;** No data available**Other Adverse Effects**

See actual entry in RTECS for complete information

**Symptoms / effects, both acute and delayed**

Product is a corrosive material. Use of gastric lavage or emesis is contraindicated.  
Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**Section 12 - Ecological Information****Ecotoxicity effects**

Reacts with water so no ecotoxicity data for the substance is available. Do not flush into surface water or sanitary sewer system.

**Persistence and Degradability****Persistence**

Miscible with water, Persistence is unlikely, based on information available.

|                                        |                                                                                                                                                            |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Bioaccumulative Potential</b>       | Bioaccumulation is unlikely                                                                                                                                |
| <b>Mobility</b>                        | The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility Highly mobile in soils |
| <b>Endocrine Disruptor Information</b> | This product does not contain any known or suspected endocrine disruptors                                                                                  |
| <b>Persistent Organic Pollutant</b>    | This product does not contain any known or suspected substance                                                                                             |
| <b>Ozone Depletion Potential</b>       | This product does not contain any known or suspected substance                                                                                             |

## Section 13 - Disposal Considerations

|                                            |                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations. |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point.                                                                                                                                                                                                                                            |
| <b>Other Information</b>                   | Chemical wastes should be disposed through a licensed commercial waste collection service. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms.             |

## Section 14 - Transport Information

### IMDG/IMO

|                                |                        |
|--------------------------------|------------------------|
| <b>UN-No</b>                   | UN1831                 |
| <b>Proper Shipping Name</b>    | SULPHURIC ACID, FUMING |
| <b>Hazard Class</b>            | 8                      |
| <b>Subsidiary Hazard Class</b> | 6.1                    |
| <b>Packing Group</b>           | I                      |

### ADG

|                                |                        |
|--------------------------------|------------------------|
| <b>UN-No</b>                   | UN1831                 |
| <b>Proper Shipping Name</b>    | SULPHURIC ACID, FUMING |
| <b>Hazard Class</b>            | 8                      |
| <b>Subsidiary Hazard Class</b> | 6.1                    |
| <b>Packing Group</b>           | I                      |

| Component                           | Hazchem Code |
|-------------------------------------|--------------|
| Sulfur trioxide<br>7446-11-9 ( 20 ) | 4WE          |

### IATA

FORBIDDEN FOR IATA TRANSPORT

|                                |                                                     |
|--------------------------------|-----------------------------------------------------|
| <b>UN-No</b>                   | UN1831                                              |
| <b>Proper Shipping Name</b>    | SULPHURIC ACID, FUMING FORBIDDEN FOR IATA TRANSPORT |
| <b>Hazard Class</b>            | 8                                                   |
| <b>Subsidiary Hazard Class</b> | 6.1                                                 |
| <b>Packing Group</b>           | I                                                   |

|                               |                                 |
|-------------------------------|---------------------------------|
| <b>Environmental hazards</b>  | No hazards identified           |
| <b>Special Precautions</b>    | No special precautions required |
| <b>Additional information</b> | None known                      |

## Section 15 - Regulatory Information

**Safety, health and environmental regulations/legislation specific for the substance or mixture****National Regulations****Australia**

See section 8 for national exposure control parameters.

**Standard for the Uniform Scheduling of Medicines and Poisons**

No poison schedule number allocated.

**Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component                         | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|-----------------------------------|-------------------------------------------------------------|------------------------|
| Sulfuric acid, fuming - 8014-95-7 | Present                                                     | -                      |
| Sulfur trioxide - 7446-11-9       | Present                                                     | -                      |

**Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

**Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

**National pollutant inventory**

Not applicable

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

**International Inventories**

| Component             | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|-----------------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Sulfuric acid, fuming | X    | X     | -         | -      | -    | -   | -    | X     | X    | X    | X     | KE-17307 |
| Sulfur trioxide       | X    | X     | 231-197-3 | -      | X    | X   | -    | X     | X    | X    | X     | KE-32690 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**International Regulations****Ozone Depletion Potential**

This product does not contain any known or suspected substance

**Persistent Organic Pollutant**

This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)**

Not applicable

**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**

Not applicable.

| Component             | CAS No    | OECD HPV | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------------------|-----------|----------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Sulfuric acid, fuming | 8014-95-7 | Listed   | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |
| Sulfur trioxide       | 7446-11-9 | Listed   | Not applicable                             | 15 tonne                                                                                  | 75 tonne                                                                                 |

**Authorisation/Restrictions according to EU REACH**

| Component             | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Sulfuric acid, fuming | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |
| Sulfur trioxide       | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

## Section 16 - Other Information

### Legend

|                                                                                                      |                                                                                                                              |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>AICS</b> - Australian Inventory of Chemical Substances                                            | <b>NZIoC</b> - New Zealand Inventory of Chemicals                                                                            |
| <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory                      | <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances |
| <b>DSL/NDSL</b> - Canadian Domestic Substances List/Non-Domestic Substances List                     | <b>ENCS</b> - Japanese Existing and New Chemical Substances                                                                  |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                     | <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                            | <b>CAS</b> - Chemical Abstracts Service                                                                                      |
| <b>TWA</b> - Time Weighted Average                                                                   | <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists                                                     |
| <b>IARC</b> - International Agency for Research on Cancer                                            | Predicted No Effect Concentration (PNEC)                                                                                     |
| <b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association | <b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code                            |
| <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships                  | <b>ADG</b> Australian Code for the Transport of Dangerous Goods by Road and Rail                                             |
| <b>NZS 5433:2012</b> - Transport of Dangerous Goods on Land                                          | <b>OECD</b> - Organisation for Economic Co-operation and Development                                                         |
| <b>LD50</b> - Lethal Dose 50%                                                                        | <b>LC50</b> - Lethal Concentration 50%                                                                                       |
| <b>EC50</b> - Effective Concentration 50%                                                            | <b>ATE</b> - Acute Toxicity Estimate                                                                                         |
| <b>WEL</b> - Workplace Exposure Limit                                                                | <b>RPE</b> - Respiratory Protective Equipment                                                                                |
| <b>DNEL</b> - Derived No Effect Level                                                                | <b>NOEC</b> - No Observed Effect Concentration                                                                               |
| <b>POW</b> - Partition coefficient Octanol:Water                                                     | <b>BCF</b> - Bioconcentration factor                                                                                         |
| <b>vPvB</b> - very Persistent, very Bioaccumulative                                                  | <b>PBT</b> - Persistent, Bioaccumulative, Toxic                                                                              |
| <b>VOC</b> - (Volatile Organic Compound)                                                             |                                                                                                                              |

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:**

|                              |                       |
|------------------------------|-----------------------|
| <b>Physical hazards</b>      | On basis of test data |
| <b>Health Hazards</b>        | Calculation method    |
| <b>Environmental hazards</b> | Calculation method    |

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.  
Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit

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and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

**Revision Date** 18-Nov-2022  
**Revision Summary** Not applicable.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**